



Generative Machine Learning Models for Data Assimilation

Sahith Edula, Mohamed Elashmony, James Evans, Juno Fang, Xuan Zhang, Ethan Chen, & Tony Deng

Mentors: Ricardo Baptista, Nisha Chandramoorthy, Giulio Trigila, & Tanya Wang

Polymath Jr.

Abstract

We study normalizing flows from the perspective of optimal transport, drawing connections between Monge's formulation, Kantorovich's formulation, and Brenier's theorem. Implementing these ideas in a practical mesh-free setting. In the quadratic cost case, Brenier's theorem ensures that the optimal transport map can be expressed as the gradient of a convex potential. Following the update rule established in the existing literature, we consider two parameterizations of this potential. The first is a radial basis function approach that departs from prior work in two key aspects: the gradient of the potential is evaluated at the original source sample point rather than at the current transported point, and convexity is enforced throughout the flow using a linear program derived from Farkas' Lemma. The second is an Input Convex Neural Network (ICNN) parameterization, which guarantees convexity by construction and therefore requires no LP-based enforcement. Experiments on synthetic two-dimensional datasets demonstrate that maintaining convexity in the potential improves stability, interpretability, and theoretical consistency of the learned maps.

Monge and Kantorovich Formulations

In optimal transport, Monge's original problem seeks a map $y : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that transports a source density $\rho(x)$ to a target density $\mu(y)$ while minimizing a cost functional

$$M(y) = \int_{\mathbb{R}^n} c(x, y(x)) \rho(x) dx.$$

This formulation enforces a one-to-one mapping between source and target points and is constrained by the change-of-variables relation $\rho(x) = J^y(x) \mu(y(x))$, where $J^y(x) = \det(\nabla y(x))$ is the Jacobian determinant. Kantorovich's relaxation replaces this rigid map with a joint distribution $\pi(x, y)$ over $\mathbb{R}^n \times \mathbb{R}^n$ that satisfies the marginal constraints

$$\int \pi(x, y) dy = \rho(x), \quad \int \pi(x, y) dx = \mu(y).$$

The problem becomes minimizing

$$\min_{\pi} \int_{\mathbb{R}^n \times \mathbb{R}^n} c(x, y) \pi(x, y) dx dy,$$

and admits a dual formulation:

$$\max_{u, v} \left[\int u(x) \rho(x) dx + \int v(y) \mu(y) dy \right]$$

subject to $u(x) + v(y) \leq c(x, y)$ for all x, y .

Brenier's Theorem

Let ρ and μ be two absolutely continuous probability densities on $\mathbb{R}^d$ with finite second moments. Then there exists a **unique** optimal transport map

$$T^* : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

pushing ρ forward to μ . Moreover, this map is the gradient of a convex potential function ϕ :

$$T^*(z) = \nabla \phi(z),$$

where $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, and $\nabla \phi(z)$ specifies the destination of the mass originally at z . Under **quadratic cost**, the optimal way to transport mass is to follow the gradient of this convex potential.

Mesh-Free Flow Implementation

We work in a mesh-free setting where the source and target densities are known only via samples $x_i \sim \rho$ and $y_j \sim \mu$ for $i = 1, \dots, m$ and $j = 1, \dots, n$. In the case of quadratic cost $c(x, y) = \frac{1}{2} \|x - y\|^2$, Monge and Kantorovich are connected via:

$$y(x) = \nabla u(x)$$

where u is the Kantorovich potential. Let $z^k(x)$ denote the iterative map at step k , initialized by $z^0(x) = x$ and updated via

$$z_i^{k+1} = \nabla u_{\beta}^k(z_i^k),$$

where

$$u_{\beta}^k(z) = \frac{1}{2} \|z\|^2 + \sum_{\ell} \beta_{\ell} F_{\ell}^k(z), \quad \nabla u_{\beta}^k(z) = z + \sum_{\ell} \beta_{\ell} \nabla F_{\ell}^k(z).$$

As the flow evolves, the map converges:

$$\lim_{k \rightarrow \infty} z^k(x) = y(x).$$

The coefficients β_{ℓ} are fitted by minimizing a Monte Carlo dual functional:

$$\hat{D}^k(\beta) = \frac{1}{m} \sum_{i=1}^m u_{\beta}^k(z_i^k) + \frac{1}{n} \sum_{j=1}^n (u_{\beta}^k)^*(y_j),$$

through gradient updates

$$\bar{\beta}_{\ell} \leftarrow \bar{\beta}_{\ell} - \eta \left[\frac{1}{m} \sum_{i=1}^m F_{\ell}^k(z_i^k) - \frac{1}{n} \sum_{j=1}^n F_{\ell}^k(y_j) \right].$$

We choose radially symmetric basis functions around centers $\tilde{z}_{\ell}$:

$$F_{\ell}(z) = r \operatorname{erf}\left(\frac{r}{\alpha}\right) + \frac{\alpha}{\sqrt{\pi}} e^{-(r/\alpha)^2}, \quad r = \|z - \tilde{z}_{\ell}\|,$$

Bandwidth α depends on local densities:

$$\alpha \propto \left(\frac{n_p}{n+m} \left(\frac{1}{\bar{\rho}(\tilde{z}_l)} + \frac{1}{\bar{\mu}(\tilde{z}_l)} \right) \right)^{1/d}.$$

This yields explicit particle updates:

$$z^{k+1} = z^k + \sum_{\ell} \beta_{\ell} f_{\ell}(r) (z^k - \tilde{z}_{\ell}), \quad f_{\ell}(r) = \frac{\operatorname{erf}(r/\alpha)}{r}$$

where the centers $\tilde{z}_{\ell}$ are re-chosen at each step to localize updates. After k steps:

$$z^k(x) = x + \text{pert}_0 + \text{pert}_1 + \dots + \text{pert}_{k-1}$$

Enforcing Convexity with Linear Programming

We study when convexity is preserved in the update

$$u_{\beta}^{k+1}(z) = u_{\beta}^k(z) + \beta_{k+1} F_{k+1}(z),$$

where u_{β}^k and F_{k+1} are convex. If all $\beta_l \geq 0$, convexity is guaranteed. When some $\beta_l < 0$, convexity must be checked explicitly. From the interpolation condition

$$u_{\beta}^{k+1}(z_0^{(j)}) \geq u_{\beta}^{k+1}(z_0^{(i)}) + g_i^{\top} (z_0^{(j)} - z_0^{(i)}) \quad \forall j,$$

convexity holds if these inequalities are feasible for g_i . By Farkas' Lemma, infeasibility of these inequalities for some i is equivalent to the existence of $\alpha \geq 0$ such that

$$\sum_{j=1}^m \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0, \quad \sum_{j=1}^m \alpha_j (u_{\beta}^{k+1}(z_0^{(j)}) - u_{\beta}^{k+1}(z_0^{(i)})) < 0.$$

Since LP solvers cannot handle strict inequalities, we instead solve

$$\min_{\alpha \geq 0} \sum_{j=1}^m \alpha_j (u_{\beta}^{k+1}(z_0^{(j)}) - u_{\beta}^{k+1}(z_0^{(i)})) \quad \text{s.t.} \quad \sum_{j=1}^m \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0.$$

If the LP is unbounded for any i , the inequalities are infeasible and u_{β}^{k+1} is not convex. In practical terms, if this condition holds, there is no affine function that acts as a valid lower local linear approximation to u_{β}^{k+1} at $z_0^{(i)}$. Geometrically, the supporting hyperplane condition fails at that point. Since any convex function can be represented as the point-wise maximum of its supporting affine functions, the absence of such a function means that no convex function can interpolate through these values at $z_0^{(i)}$.

Divergence from Published Literature

Our method departs from the published literature in two functional ways. The first, we use ICNN components to enforce convexity, parameterizing the potential as

$$T(x) = \nabla \phi_{\alpha}(x) = \alpha x + \sum_{k=1}^K \beta_k \nabla F_k(x),$$

with each F_k implemented as an *Input Convex Neural Network (ICNN)*. The second, we use a special localized elementary map while enforcing convexity with LP:

$$X_i^{n+1} = X_i^n + B \nabla_{X^0} F(r_i^{n+1}) = x_i^n + \beta f(r_i^{n+1}) (x_i^0 - \hat{x}^{n+1})$$

where

$$r_i^{n+1} = \|X_i^0 - \hat{X}^{n+1}\|, \quad f(r) = \frac{1}{r} \frac{dF(r)}{dr}.$$

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